Scientific Committee on Consumer Safety

SCCS

SCIENTIFIC ADVICE

on the safety of Phytonadione Epoxide

(Vitamin K1 oxide)

(CAS No. 25486-55-9, EC No. 247-022-9)

The SCCS adopted this document
by written procedure on 4 November 2026

ACKNOWLEDGMENTS

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1. ABSTRACT

The SCCS concludes the following:

In light of all data submitted during the call for data:

1. *Does the SCCS consider Phytonadione Epoxide (Vitamin K1 oxide) safe when used as an ingredient in cosmetic products up to a maximum concentration of 1%?*

The SCCS considers Phytonadione Epoxide (Vitamin K1 oxide) not safe when used in cosmetic products at a concentration up to 1%. There is clear evidence from several case reports in published literature and SUEs that Phytonadione Epoxide can cause serious allergic skin reactions that often require medical treatment.

2. *Alternatively, what is according to the SCCS, the maximum concentration considered safe for Phytonadione Epoxide (Vitamin K1 oxide) in cosmetic products?*

Due to the absence of quantitative data to derive a Point-of-Departure (PoD) for toxicological hazards, including skin sensitisation, and the lack of information regarding consumer exposure, the SCCS has not been able to complete a safety assessment on Phytonadione Epoxide. It is therefore not possible for the SCCS to determine a safe maximum concentration for Phytonadione Epoxide in cosmetic products.

3. *Does the SCCS have any further scientific concerns regarding the use of Phytonadione Epoxide (Vitamin K1 oxide) in cosmetic products?*

No data on mutagenicity/genotoxicity were provided, either during the call for data or in the published literature. Therefore, the potential mutagenicity/genotoxicity of Phytonadione Epoxide cannot be excluded without the provision of relevant experimental data.

Keywords: SCCS, scientific advice, Phytonadione Epoxide – Vit.K1 oxide, Regulation 1223/2009, CAS No. 25486-55-9, EC No. 247-022-9

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Two independent non-food Scientific Committees provide the Commission with the scientific advice it needs when preparing policy and proposals relating to consumer safety, public health and the environment. The Committees also draw the Commission's attention to the new or emerging problems, which may pose an actual or potential threat.
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In addition, the Commission relies upon the work of the European Food Safety Authority (EFSA), the European Medicines Agency (EMA), the European Centre for Disease prevention and Control (ECDC) and the European Chemicals Agency (ECHA).

SCCS

The Committee shall provide Opinions on questions concerning health and safety risks (notably chemical, biological, mechanical and other physical risks) of non-food consumer products (for example cosmetic products and their ingredients, toys, textiles, clothing, personal care and household products such as detergents, etc.) and services (for example: tattooing, artificial sun tanning, etc.).

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2. MANDATE FROM THE EUROPEAN COMMISSION

Background

Regulation (EC) No 1223/2009 on cosmetic products lays down that cosmetic products made available on the Union market must be safe for human health when used under normal or reasonably foreseeable conditions of use. In this framework, the European Commission relies on the independent scientific advice of the Scientific Committee on Consumer Safety (SCCS) for the assessment of substances used in cosmetics that may raise concerns for consumer health.

Phytonadione Epoxide (INCI) (Vitamin K1 oxide) (CAS No. 25486-55-9, EC No. 247-022-9) is a synthetic or naturally-occurring oxidised form of vitamin K1 (phytonadione), resulting from the epoxidation of the naphthoquinone ring. In cosmetic products, it is mainly used as a skin conditioning agent and is reported to contribute to reducing the appearance of dark circles and redness, as well as improving skin tone uniformity. It is therefore used in products such as eye contour creams, anti-ageing serums and formulations intended for sensitive or redness-prone skin, among others

Phytonadione (Vitamin K1) was previously used in cosmetic products for skin conditioning purposes. However, due to reported cases of allergic contact dermatitis associated with its topical use, Vitamin K1 was prohibited for use in cosmetic products in the European Union in 2009 (see entry 1371 in Annex II to the Cosmetics Regulation). Following this restriction, Phytonadione Epoxide was introduced as an alternative ingredient in cosmetic formulations.

In recent years, concerns have been raised regarding the sensitisation potential of Phytonadione Epoxide. In particular, several cases of allergic contact dermatitis, including severe skin reactions and photo-induced responses, have been reported in relation to cosmetic products containing this substance, especially eye contour products. Reports from dermatological centres, including cases identified in Spain, indicate an increasing number of sensitisation cases linked to exposure to Phytonadione Epoxide. Furthermore, published literature suggests that diagnosis may be difficult using standard patch testing concentrations, potentially leading to under-recognition of sensitisation cases.

Given the possible health risks related to its sensitisation potential, and considering the use of Phytonadione Epoxide in leave-on cosmetic products intended for use on sensitive skin areas, the Commission considered it necessary to collect comprehensive data on the safety of this substance when used in cosmetic products.

To obtain a clearer understanding of the exposure conditions, use patterns and toxicological profile of Phytonadione Epoxide, the Commission launched a call for data between May 2025 and May 2026, which sought to gather comprehensive scientific information on the safety of Phytonadione Epoxide when used in cosmetic products.

Based on the information submitted through the call for data, together with evidence from scientific literature, the Commission requests the SCCS to carry out a safety assessment of this ingredient, considering the available evidence, and determining whether consumer exposure under realistic conditions of use is safe for all relevant age groups.

Terms of reference

In light of all data submitted during the call for data:

1. *Does the SCCS consider Phytonadione Epoxide (Vitamin K1 oxide) safe when used as an ingredient in cosmetic products up to a maximum concentration of 1%?*

2. *Alternatively, what is, according to the SCCS, the maximum concentration considered safe for Phytonadione Epoxide (Vitamin K1 oxide) in cosmetic products?*

3. *Does the SCCS have any further scientific concerns regarding the use of Phytonadione Epoxide (Vitamin K1 oxide) in cosmetic products?*

3. SCIENTIFIC ADVICE

3.1 CHEMICAL AND PHYSICAL SPECIFICATIONS

3.1.1 Chemical identity

3.1.1.1 Primary name and/or INCI name

Phytonadione epoxide

Refs. Cayman chemical; consulted on 30-7-2026 (www.caymanchem.com/product/16017/vitamin-k1-2-3-epoxide); Pubchem; consulted on 30-7-2026 (pubchem.ncbi.nlm.nih.gov/compound/5280845

3.1.1.2 Chemical names

IUPAC name: 7a-methyl-1a-[(E)-3,7,11,15-tetramethylhexadec-2-enyl]naphtho[2,3-b]oxirene-2,7-dione

Synonyms: 2,3-epoxy Phylloquinone, (2,3-Epoxyphytyl)menaquinone, Phylloquinone oxide, Vitamin K oxide, Vitamin K1 2,3-epoxide

Ref. Cayman chemicals, Pubchem

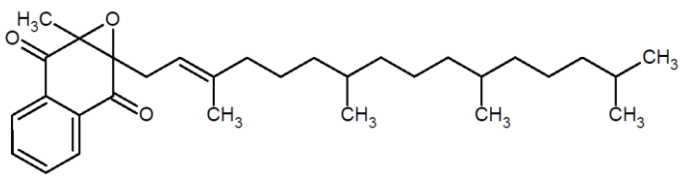
3.1.1.3 Trade names and abbreviations

3.1.1.4 CAS / EC number

CAS number: 25486-55-9
EC number: 247-022-9

Ref. Pubchem

3.1.1.5 Structural formula



Ref. Pubchem

3.1.1.6 Empirical formula

$C_{31}H_{46}O_3$

Ref. Pubchem

3.1.2 Physical form

3.1.3 Molecular weight

3.1.4 Purity, composition and substance codes

3.1.5 Impurities / accompanying contaminants

3.1.6 Solubility

3.1.7 Partition coefficient (Log Pow)

3.1.8 Additional physical and chemical specifications

Where relevant:
- organoleptic properties (colour, odour, taste if relevant)
- melting point
- boiling point
- flash point
- vapour pressure
- density
- viscosity
- pKa
- pH
- refractive index
- UV/visible light absorption spectrum

3.1.9 Homogeneity and Stability

/

SCCS comment
The SCCS notes that commercially available Vitamin K1 oxide is recommended for storage at −20°C. Appropriate data on purity, impurities, potential degradation products, homogeneity, and stability under relevant storage and use conditions have not been provided during the call for data.

Ref: https://www.sigmaaldrich.com/GR/en/documents/spec/19697/SIGMA

3.2 TOXICOKINETICS

3.2.1 Dermal / percutaneous absorption

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3.2.2 Other studies on toxicokinetics

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SCCS comment
Data on dermal absorption studies were neither provided during the public call for data, nor retrieved in the literature search conducted by the SCCS. A default dermal absorption of 50% is therefore used in this assessment.

3.3 EXPOSURE ASSESSMENT

3.3.1 Function and uses

Taken from information provided by Respondent 1 and 2
According to reports provided by Respondents to the public call for data, Phytonadione Epoxide (Vitamin K1 oxide) is an oxidized derivative of vitamin K1. This compound forms when the hydroquinone is oxidized to 2,3-epoxide by the gamma-glutamyl carboxylase (GGCX) enzyme. It differs from vitamin K1 structure by the epoxidation of the naphthoquinone ring. The substance is used as a cosmetic ingredient with skin conditioning properties, contributing to the improvement of the appearance of skin affected by redness and visible capillaries. Phytonadione epoxide is used in a leave-on facial cosmetic product intended for adult consumers, applied once daily to facial skin. The eye area should be avoided.

3.3.2 Calculation of SED/LED

Taken from information provided by Respondent 1

A Cosmetic Product Safety Report (CPSR) provided by Respondent 1 calculated the SED for a leave-on facial cream intended for topical application and which is not rinsed off after application (retention factor=1). The product is applied by gently rubbing a thin layer onto clean facial skin, avoiding the eye area and mucous membranes. Exposure assessment indicates a daily use of approximately 1,54 g of product applied to facial skin. The exposure route is dermal, and systemic exposure is expected to be low under normal conditions of use.

For Phytonadione Epoxide, the SED was calculated using the maximum concentration in cosmetics of 1% and a dermal absorption value of 50%. This resulted in a SED of 0.0642 mg.

Table 1: Exposure to the cosmetic product (CPSR, 2024)

Application place:	Half of the head area, the skin surface of the neck and décolletage
Population or exposed:	adults
Skin surface area of application:	965cm² - the area determined according to the SCCS guidelines for the average consumer was used for the calculation
Average body weight of an average person exposed to the product:	60 kg
Routes of exposure:	dermal
Routes of secondary exposure:	unlikely
The quantities of product used:	1.54 g/day - according to SCCS recommendations
Frequency of application:	once a day, at night - according to the manufacturer's recommendations
Product type:	„leave-on" product
Retention Factor:	1.0

(1) SCCS/1647/22
(2) RIVM report 320104001/2006

Ref. CPSR, 2024

3.4 TOXICOLOGICAL EVALUATION

3.4.1. Irritation and corrosivity

3.4.1.1 Skin irritation

/

3.4.1.2 Mucous membrane irritation / eye irritation

/

3.4.2 Skin sensitisation

Taken from information provided by Respondent 2

ARE-Nrf2 luciferase assays KeratinoSens™

Guideline: OECD TG442D
Test System: Keratinosens cell line
Replicates: Two
Test substance: Vitamin K1 oxide
Batch: 3127/30011L
Purity: ≥99%
Solvent: DMSO
Dose level:
Treatment period: 48 hours
Positive control: Cinnamic Aldehyde (CAS 14371-10-9; ≥ 98% purity)

Negative control: Solvent (1% DMSO in culture medium)
GLP: not reported

Methodology
KeratinoSens cells were exposed to 12 concentrations of Vitamin K1 Oxide for 48 hours. Additionally, cells were treated with cinnamic aldehyde as positive control for skin sensitization. After the incubation period, samples were processed for luciferase activity by measuring luminescence and cell viability was analysed by the MTT assay. Two independent runs were performed. Data were statistically analysed.
Test chemicals are considered positive in the KeratinoSens™ test method if they induce a statistically significant induction of the luciferase activity above a given threshold (i.e. ≥ 1.5 fold, or 50% increase), below a defined concentration which does not significantly affect cell viability (i.e. below 1000 µM and at a concentration at which the cellular viability is above 70%).

Results
Vitamin K1 oxide did not induce luciferase activity ≥1.5 fold at cell viabilities higher than (>) 70%. Cinnamic aldehyde was positive.

Conclusion
Under the tested conditions following OECD 442D guidelines "In vitro Skin Sensitization", treatment with Vitamin K1 Oxide is negative for skin sensitization.

SCCS comment
The available human data (see 3.4.9.) provide clear evidence that Phytonadione Epoxide can cause severe allergic contact dermatitis in humans.
Although Phytonadione Epoxide is reported as negative in the KeratinoSens® assay, as described in the SCCS Notes of Guidance (12th revision), skin sensitisation potential and potency cannot be evaluated using a standalone new approach method (NAM). NAMs covering different key events of the skin sensitisation Adverse Outcome Pathway are required, either in Defined Approaches to Testing and Assessment of skin sensitisation (DASS), or as part of an integrated approach, such as the Next-Generation-Risk-Assessment (NGRA) for skin sensitisation.

3.4.3 Acute toxicity

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3.4.3.1 Acute oral toxicity

3.4.3.2 Acute dermal toxicity

3.4.3.3 Acute inhalation toxicity

3.4.4 Repeated dose toxicity

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3.4.4.1 Repeated dose (28 days) oral / dermal / inhalation toxicity

3.4.4.2 Sub-chronic (90 days) oral / dermal / inhalation toxicity

3.4.4.3 Chronic (> 12 months) toxicity

3.4.5 Reproductive toxicity

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3.4.5.1 Fertility and reproduction toxicity

3.4.5.2 Developmental Toxicity

3.4.6 Mutagenicity / genotoxicity

/

3.4.6.1 Mutagenicity / genotoxicity *in vitro*

3.4.6.2 Mutagenicity / genotoxicity *in vivo*

3.4.7 Carcinogenicity

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3.4.8 Photo-induced toxicity

3.4.8.1 Phototoxicity / photo-irritation and photosensitisation

Taken from information provided by Respondent 2

***In vitro* phototoxicity study**

Guideline:	OECD 432 3T3 Neutral Red Uptake (NRU) Phototoxicity assay
Test System:	Balb/c 3T3 cell line
Replicates:	not reported
Test substance:	Vitamin K and Vitamin K1 oxide
Batch:	9191-A1 (Vitamin K) and 3127/40017(vitamin K oxide)
Purity:	not reported
Solvent:	DMSO
Dose level:	45, 15, 5, 1.67, 0.56, 0.18, 0.062, 0.020 μg/mL
Treatment period:	1 hour
Positive control:	Chlorpromazine (CPZ), CAS 69-09-0
Negative control:	Solvents of test item (DMSO) and positive reference (ethanol)
GLP:	not reported

Methods

The day before the exposition to the test item, Balb/c 3T3 cells were trypsinized and counted. A suspension of 50,000 cells/mL was prepared and 200 µL of the suspension were added to each well in 96-well plates (10,000 cells/well), except in the peripheral wells 3T3 cells were treated with each product at eight different concentrations for 1 hour. Thereafter, half of the cells were exposed to simulated solar irradiation, whereas the other half was kept in the dark. Cell viability was calculated after 24 hours of incubation by means of the neutral red assay.

Chlorpromazine (CPZ; CAS 69-09-0) was used as a positive reference item. For this reference item, 8 concentrations were prepared using ethanol as solvent: 200, 66.67, 22.22, 7.407, 2.47, 0.823, 0.274, and 0.091 µg/mL. Solvent of the test item or the positive reference item (DMSO and ethanol respectively) were used as negative controls.
Vitamin K (batch 9191-A1) and Vitamin K1 oxide (batch 3127/40017) were used as a test items.
In accordance with OECD Guideline 432, the maximum concentration to be tested in a phototoxicity assay should not exceed 1 mg/mL. However, due to the physicochemical characteristics of the test items, dissolution in aqueous solvents was not possible. Consequently, dimethyl sulfoxide (DMSO) was employed as an alternative solvent.
Preliminary solubility testing demonstrated that both test items were fully soluble in DMSO at a concentration of 4.5 mg/mL. As the DMSO concentration must be limited to prevent solvent-induced cytotoxicity in the cell system, the highest concentration tested in the assay was set at 45 µg/mL. From this concentration, seven serial dilutions were subsequently prepared for testing in 1/3 dilutions.

3T3 cells were seeded at 10,000 cells/well in 96-well plates. Cells were exposed for 1 hour at 37°C and 5% CO2. Two 96-well plates were prepared for each test item and reference control. One plate of each condition (+IRR) was irradiated for 42 minutes at 5 J/cm2 while the other plates (-IRR) were kept in the dark for the same amount of time. Then, culture media was removed, cells were washed and 100 µL of neutral red were added to each well. Cells were incubated for 3 hours at 37°C and 5% CO2 and then were washed again. Finally, the neutral red was solubilized by adding 100 µL to each well of a solution of Acetic acid/Ethanol. Plates were incubated 10 minutes at room temperature and absorbance at 540 nm was measured.

Results

According to OECD Guideline 432, if an IC50 value cannot be determined, the Photo Irritation Factor (PIF) cannot be calculated for that test item. In such cases, the Mean Photo Effect (MPE) is used as an alternative parameter. In this study, the IC50 value could not be determined for non-irradiated cells exposed to Vitamin K1, as well as for both irradiated and non-irradiated cells exposed to Vitamine K1 Oxide, since no viability values below 50% were observed. Consequently, the MPE and PIF values for the controls and test items were calculated using a software tool developed within the OECD 432 framework.

To predict the phototoxic potential, the concentration responses obtained in the presence (+Irr) and in the absence (-Irr) of solar irradiation were compared and photo-irritation factor (PIF) and/or the mean photo effect (MPE) was calculated:

Table 2: PIF (and MPF) calculations values.

Sample	PIF $\frac{IC50\ (-Irr)}{IC50\ (+Irr)}$	MPE $\frac{\sum_{i=1}^{n} W_i\ PE_{ci}}{\sum_{i=1}^{n} W_i}$
Vitamin K1	Not available	0.477
Vitamin K1 Oxide	Not available	-0.023
CPZ	83.029	0.569

Conclusion
The test item Vitamin K1 was classified as phototoxic under the conditions assayed.
The test item Vitamin K1 Oxide was classified as non-phototoxic under the conditions assayed.

3.4.8.2 Photomutagenicity / photoclastogenicity

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3.4.9 Human data

Taken from information provided by Respondent 2

Single patch test (human)
A cutaneous compatibility study conducted in 10 healthy adult volunteers (male and female, aged 18 to 70 years) showed no evidence of skin irritation following a single occlusive application of 0.02 g of the test substance (vitamin K1 oxide) over a 50 mm² area for 48 hours.

Conclusions on skin irritation from Respondent 2
Vitamin K1 oxide demonstrated no evidence of skin irritating properties in an *in vivo* human patch test in 10 healthy adult volunteers. Moreover, it has extensively been used in cosmetic products without significant irritation reactions reported.
Vitamin K1 oxide is concluded to show low potential for skin irritation at the concentration levels used in cosmetic formulations.

SCCS comment
The full study report of this single patch test was not provided by the respondent. In the absence of *in vitro* skin irritation data, a patch test in 10 healthy volunteers is not sufficient to rule out the skin irritation potential of Phytonadione Epoxide.

Taken from information provided by Respondent 1

Human Repeat Insult Patch Test (HRIPT)

During the public call for data, a HRIPT report was provided. In short, 50 healthy male and female volunteers aged 18-70 years were enrolled in the study. The HRIPT was performed with a vitamin K cream. No irritation or sensitisation reactions were found under tested conditions.

SCCS comment

In several previous Opinions, the SCCS has expressed ethical concerns about human skin sensitisation tests, including the HRIPT. In addition, the HRIPT was conducted in a small group of volunteers using a product and not the cosmetic ingredient itself. The specific ingredients in the investigated product, and the concentration of vitamin K, were not specified in the submission. Therefore, the SCCS could not use this study in support of the safety of Phytonadione Epoxide.

Taken from information provided by Respondent 2

A table summarising human data on skin reactions together with patch test results for several products was provided by Respondent 2 (Table 4).

Although different, the lesions observed in all cases were compatible with allergic contact dermatitis. Patch test positivity was described to be dose-dependent (Aerts, O., *et al.*, 2012; Gatica-Ortega, M. E., *et al.*, 2023).

Case reports indicate that patients previously sensitized to vitamin K1 may also react to its oxide form, suggesting a possible cross-reactivity between the two substances (Foti, C., *et al.*, 2022; Uter, W., *et al.*, 2020).

Table 3: Human data

Patients	Product	Skin reactions	Concentration for Patch Test	Results of Patch Test	Reference
♀47 y.o. ♀63 y.o. ♀51 y.o.	Auriderm XO®, Auriga	Pruritic lesions on their legs, lesions on her right breast	*(for the ♀47 y.o.)* Auriderm and VK1O 1% pet.	Positive	García-Gavín, J., et al., 2010
NA	Auriderm XO®	Localized dermatitis that eventually generalized	Auriderm, Konakion 10mg/ml (1% VK1) and Konakion 1 mg/ml (0.1% VK1)	Auriderm ++ Konakion 10mg/ml ++ Konakion 1mg/ml +	Aerts, O., et al., 2012
♂6 y.o. ♀35 y.o.	Cream containing vitamin K1 oxide (Arnika gel)	Widespread erythematous eruption (boy) and erythema multiforme-like eruption following a mesotherapy procedure (woman)	VK1O 5% pet. and Arnika gel	Positive (both patients)	Pastor-Nieto, M.A., et al., 2017
♀57 y.o.	Eye correct Platinum and Konakion	Acute eczema in the eye	Eye correct Platinum, VK1O 1% pet, VK1O 5% pet., and Konakion (1% VK1)	Positive for cream 'as is', to vitamin k1 oxide, and to vitamin K1	Schneller-Pavelescu, L. et al., 2018
17 cases (all♀, $\bar{x}$ = 48.17 y.o.) [review cases, 2019-2023)	Vitamin K1 oxide	Allergic contact dermatitis (More than half were caused by one eye-contour product, K-Ox Eyes, Isdin)	1, 5, 10, 20 % VK1O	Positive in 6/8 patients (75%) at 1%, 13/14 (92.9%) at 5%, and in all patients at both 10% (11/11) and 20% (8/8)	Gatica-Ortega, M. E., et al., 2024
♀39 y.o	Facial cream containing vitamin K1 oxide and bisabolol	Intense pruritus and oedematous eczema on her face, neck, and neckline	Vitamin K1 oxide at 1% and 5%	Positive	de la Rosa-Fernández, E., et al., 2024

Cross reactivity

Cross-reactions among vitamin K isoforms have been described:
For example, in 1999, a 40-year-old woman developed a cutaneous allergic reaction to intramuscular vitamin K1. Investigation by patch and intradermal testing revealed delayed-type hypersensitivity to vitamin K1 as well as vitamin K3 (Wong, DA. & Freeman, S., 1999).

In 2002, the day after intramuscular injection of vitamin K1, a 27-year-old-woman developed a relapsing and remitting eczematous reaction localized to the injection site. On patch testing, she was positive to vitamin K1 and cross-reacted to vitamin K4 (Sommer, S., *et al.*, 2002).

Cross-reactivity between vitamin K1 and its oxide has also been reported, based on the results of a patient who presented with ACD caused by a cosmetic containing vitamin K1 oxide. As the manufacturer did not provide the allergen, the patient was patch tested with the non-oxidized vitamin K1 form, to which he developed a positive reaction (Aerts, O., *et al.*, 2012; Pastor-Nieto, MA., *et al.*, 2017; Schneller-Pavelescu, L., *et al.*, 2018).

Conclusions of Respondent 2 on skin sensitisation

Overall, after revision of available data for the toxicological assessment of the potential risk of skin sensitization or photosensitization for vitamin K1 oxide, it is concluded that not

enough scientific evidence is available to completely rule out the risk of skin sensitization or photosensitization for the target compound. However, available data suggest at least a reduced potential of photosensitization for Vitamin K1 oxide compared to Vitamin K1.

SCCS comment

The experimental data provided by Respondent 2 are not sufficient to be able to conclude on the skin sensitisation potential and potency of Phytonadione Epoxide. Human data, as shown in Table 3, clearly demonstrate that Phytonadione Epoxide can induce allergic contact dermatitis. In the absence of any experimental data to estimate the potency of Phytonadione Epoxide, it is not possible to conclude that Phytonadione Epoxide has a lower potential of (photo)sensitisation compared to vitamin K.

Taken from information provided by Respondent 3

Respondent 3 provided the SCCS with several publications on allergic contact dermatitis caused by Phytonadione Epoxide. Some of these publications are summarised in Table 3, provided by Respondent 2. An additional literature search conducted by the SCCS did not result in any publications that were not already provided by these Respondents.

In short:

- Three cases of allergic contact dermatitis caused by the same cosmetic product containing vitamin K1 oxide were reported. Patients presented with perifollicular papules or pruritic lesions on skin exposed to the cream with vitamin K oxide. Patch testing with the product resulted in positive patch test results in all patients. Two patients were also patch tested with vitamin K oxide, with positive results in both cases (Garcia-Gavin *et al*, 2010).
- Pastor-Nieto *et al.* (2017) described two case reports with symptoms of erythema multiforme-like reactions, an uncommon non-eczematous clinical variant of allergic contact dermatitis. In these cases, erythematous papules developed on skin areas where a gel vitamin K1 oxide was used. Positive results were obtained in the patch test by the product and by vitamin K oxide. This shows that vitamin K oxide was the cause of the allergic reactions in both cases.
- A case report describes cross-reactivity between non-oxidized and oxidized vitamin K1. A woman developed acute eczema after application of eye contour cream with vitamin K1 oxide. Positive results were obtained in patch tests with the cream, phytonadione epoxide and Konakion, i.e. phytomenadione (vitamin K1). Thirteen years earlier, she had similar symptoms after using an eyelid cream with vitamin K1. At that time the allergy was confirmed with positive reactions in the patch test with the product, but not with vitamin K itself. According to the authors, this case report demonstrates cross-reactivity between vitamin K and vitamin K oxide (Schneller-Pavelescu *et al.*, 2018)
- Severe allergic contact dermatitis was reported in a case report. A female that used a face cream with vitamin K oxide developed pruritic oedematous eczema affecting her entire face and neck. She was treated in the emergency room with prednisone, but lesions remained intense for five days before showing improvement. Patch testing was positive for both bisabolol and phytonadione epoxide (Rosa-Fernandez *et al.*, 2024).
- In a Spanish study, records of 20 patients with ACD to cosmetics were reviewed between January 2019 and June 2023. Phytonadione epoxide gave positive patch test or repeated open application test (ROAT) reactions in 17 patients. All presented with symptoms on the eyelids. Ten of the 17 patients required medical attention at the emergency department and/or oral corticosteroids.

Undesirable and Serious undesirable effects (UE and SUE)

During the public call for data, information on (serious) undesirable effects (UEs and SUEs) in humans were provided. They are summarised below.

Taken from information provided by Respondent 4

A presentation was provided that summarized UE and SUEs related to Phytonadione epoxide topical exposure from Spain and Portugal. Figure 1 shows SUEs reported in Spain. Since 2016 a gradual increase of SUEs, both non-serious and serious, has been observed in Spain. All SUEs were on skin effects, e.g. aspecific dermatitis and allergic contact dermatitis. The latter was often confirmed in a patch test to the product with vitamin K oxide.

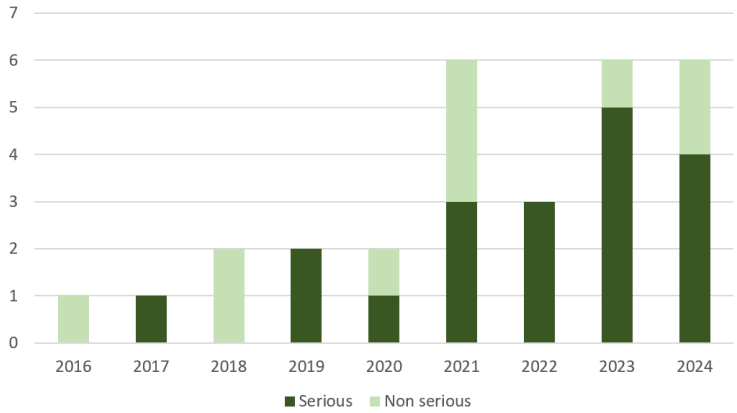


Figure 1: Sue and UE related to products containing vitamin K epoxide in Spain and Portugal.

The main concerns of Respondent 3 were that products with vitamin K oxide are gaining popularity. The Respondent also expressed concerns that due to underreporting, there could be missing cases and that cases that do not fall within the seriousness criteria are not notified. Finally, cross-sensitisation to vitamin K1, which is not allowed to be used in cosmetics in the EU due to the risk of allergic contact dermatitis, is a concern. Not only in persons sensitised via cosmetics. Vitamin K1 is also an important therapeutic agent. In cases of pre-existing sensitisation acquired by topical application of Vitamin K1 in cosmetics, an individual may experience an allergic reaction to Vitamin K1 treatment and/or might not be able to receive Vitamin K1 therapy should it be required (SCCS/1313/10).

Taken from information provided by Respondent 1

List of adverse reactions from the market (07.2018-01.2024)/ side effect reported from the market:

- 18.04.2023 - Severe redness of facial skin after application of vitamin K cream and eye cream. Greater likelihood of irritation reaction after Vit. K cream. - classified as very likely
- 07.11.2022 - Very severe allergic reaction of the face, neck and decolletage after applying a sample of the cream. Intensive dermatological treatment, the lesions have persisted for a week, treatment lasted at least another 2 weeks. - classified as very likely
- From 2012-2018: 13 adverse reactions reported, including 1 severe adverse reaction in 2016 (see Table below).

Table 4: Adverse reactions reported

17.04.2016	Allergy	classified as likely
23.05.2016	Allergy	classified as likely
02.08.2016	Irritation after cosmetics	classified as likely
2014.12.12	Allergy	classified as likely
2014.12.12	Allergy	classified as likely
2015.04.29	Irritation after cosmetics	classified as likely
2015.04.24	Irritation after cosmetics	classified as likely
2016.03.25	Irritation after cosmetics	classified as likely
2017.01.09	Irritation after cosmetics	classified as likely
2017.01.26	Allergy	classified as likely
2017.06.30	Irritation after cosmetics	classified as likely
2017.09.06	Irritation after cosmetics	classified as likely
2018.06.07	Irritation after cosmetics	classified as likely

The effects described in Table 4 did not require ambulatory treatment. They disappeared after discontinuation of the product.

A detailed description of severe reaction was described in the report from Respondent 1. In short, summarized by the SCCS:

A women experienced serious allergic reactions after using products containing Phytonadione Epoxide. After applying the cream to her face and neck, she experienced itching and swelling in her face and neck, like hives. A day later the lesions became very severe, a very large erythema developed on the cheeks, face, neck, forehead, temples and behind the ear, general swelling, blisters, itching, burning, feeling of hot skin. She required medical attention. Later, the symptoms increased, a very large swelling of the left half of the face developed, swelling of the eyes, huge blisters on the neck, lesions heavily red, facial erythema, itching, burning, swelling of the ear, muffled hearing, swelling in the throat. She required medical attention again and received new medications. Later she also went to the emergency department to receive additional treatment, because of persistence of the shortness of breath and swelling in her ears and throat. Treatment gradually improved the symptoms, but facial skin is still sensitive to heat. A diagnostic patch test confirmed that the vitamin K cream caused the allergic symptoms.

In the period 01/08/2016-31/06/2017, two cases of adverse reaction to a cosmetic product after the change of the package insert text were recorded. The complaints were reported in January 2017. Both cases required medical intervention. Customers had a history of not following the warning on the package. One reaction was confirmed in the interview, the other was not - no feedback from the Consumer.

SCCS overall comment on allergic contact dermatitis

In recent years, there have been several reports on Serious Undesirable Effects (SUEs) on cosmetic ingredients that contained Phytonadione epoxide. In addition, several publications have described case reports on allergic skin reactions. Severe symptoms have been reported in these publications that often required medical attention and treatment. Positive patch tests results showed that the allergic reactions were caused by Phytonadione Epoxide, demonstrating its skin sensitising potential. In the absence of experimental skin sensitisation data, with the exception of the result of a Keratinosens® test, it is not possible to compare the potency of Phytonadione epoxide with vitamin K1.

The SCCS also has concerns about the possible immunological cross-reactivity with Vitamin K1, a cosmetic ingredient that was forbidden in the European Union after several SUEs on

severe allergic contact dermatitis (SCCS/1313/10). Persons sensitised to vitamin K may react with (severe) allergic reactions to Phytonadione Epoxide due to cross-reactivity. Immediate-type allergic skin reactions have been documented for Vitamin K1 (Gimenez-Arnau 2005). It is not known whether the delayed-type cross-reactivity between Vitamin K1 and its oxidized form (Phytonadione epoxide) can also carry a risk for the more hazardous systemic immediate allergic reactions to Phytonadione Epoxide.

Another important concern is that Vitamin K1 is also an important therapeutic agent. In cases of pre-existing sensitisation acquired by topical application of Phytonadione Epoxide in cosmetics, an individual may experience an allergic reaction to Vitamin K1 treatment, and/or might not be able to receive Vitamin K1 therapy, should it be required.

3.4.10 Special investigations

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3.5 SAFETY EVALUATION (INCLUDING CALCULATION OF THE MoS)

Both the public call for data and the literature search conducted by the SCCS resulted in a limited number of reports and scientific publications. These were all on allergic contact dermatitis or phototoxicity. No data was available or provided on other toxicological endpoints. Hence, there was no data to derive a Point-of-Departure (PoD) for Phytonadione Epoxide. Therefore, it is not possible to perform a safety assessment deriving a margin of safety for the use of Phytonadione Epoxide in cosmetic products

3.6 DISCUSSION

Exposure

No dermal absorption study was provided during the public call for data nor was it retrieved in the literature search conducted by the SCCS. A dermal absorption of 50% is used in this case.

Toxicological Evaluation

Skin sensitisation
Phytonadione Epoxide was negative in the Keratinosens™ under the conditions tested. The experimental data are not sufficient to conclude that Phytonadione Epoxide is not a skin sensitiser. Available human data provide clear evidence that Phytonadione can cause severe allergic contact dermatitis in humans.

Photo-induced toxicity
Phytonadione Epoxide was not phototoxic in the Neutral Red Uptake (NRU) Phototoxicity assay under the conditions tested.

Human data
Several publications show that Phytonadione Epoxide can cause allergic contact dermatitis. A case report also shows cross-reactivity between Phytonadione Epoxide and Vitamin K1. In addition, information on (serious) undesirable effects (UEs and SUEs) was provided to the SCCS during the public call for data. Allergic reactions were sometimes severe and required

medical attention and treatment. There is insufficient experimental data to compare the potency of Phytonadione Epoxide with Vitamin K.
The SCCS has concerns about cross-reactivity with Vitamin K1, an ingredient that was forbidden in the European Union after several SUEs on severe allergic contact dermatitis (SCCS/1313/10). People sensitised earlier to Vitamin K may react with (severe) allergic reactions to Phytonadione Epoxide due to cross-reactivity.
Another important concern which SCCS wants to reiterate, is that Vitamin K1 is an important therapeutic agent. In cases of pre-existing sensitisation acquired by topical application of Phytonadione Epoxide in cosmetics, an individual may experience an allergic reaction to Vitamin K1 treatment and/or might not be able to receive Vitamin K1 therapy should it be required.

Special investigation
The public call for data and the literature search conducted by the SCCS resulted in a limited number of reports and scientific publications. These were all on allergic contact dermatitis or phototoxicity. No data was available or provided on other toxicological endpoints. Hence, there was no data to derive a Point-of-Departure (PoD) for Phytonadione epoxide. Therefore, it is not possible to perform a safety assessment deriving a margin of safety for the use of Phytonadione Epoxide in cosmetic products.

4. CONCLUSION

In light of all data submitted during the call for data:

1. *Does the SCCS consider Phytonadione Epoxide (Vitamin K1 oxide) safe when used as an ingredient in cosmetic products up to a maximum concentration of 1%?*

The SCCS considers Phytonadione Epoxide (Vitamin K1 oxide) not safe when used in cosmetic products at a concentration up to 1%. There is clear evidence from several case reports in published literature and SUEs that Phytonadione Epoxide can cause serious allergic skin reactions that often require medical treatment.

2. *Alternatively, what is according to the SCCS, the maximum concentration considered safe for Phytonadione Epoxide (Vitamin K1 oxide) in cosmetic products?*

Due to the absence of quantitative data to derive a Point-of-Departure (PoD) for toxicological hazards, including skin sensitisation, and the lack of information regarding consumer exposure, the SCCS has not been able to complete a safety assessment on Phytonadione Epoxide. It is therefore not possible for the SCCS to determine a safe maximum concentration for Phytonadione Epoxide in cosmetic products.

3. *Does the SCCS have any further scientific concerns regarding the use of Phytonadione Epoxide (Vitamin K1 oxide) in cosmetic products?*

No data on mutagenicity/genotoxicity were provided, either during the call for data or in the published literature. Therefore, the potential mutagenicity/genotoxicity of Phytonadione Epoxide cannot be excluded without the provision of relevant experimental data.

5. MINORITY OPINION

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6. REFERENCES

Provided by Respondent 1:

- Submission from Respondent 1. Summary of supporting data (2026). Submitted to the SCCS during the public call for data.
- Cosmetics Product Safety Report (CPSR) (2024)

Provided by Respondent 2:

- *In vitro* study (OECD 432) for two products: vitamin K1 oxide and vitamin K1.
- Vitamin K1 oxide toxicological assessment of potential risk of skin sensitization.
- Analysis of skin sensitization (ARE-Nrf2 luciferase test method). Final report.

Literature provided by Respondents 2 and 3:

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11. Wong, D. A. & Freeman, S. Cutaneous allergic reaction to intramuscular vitamin K1. Australasian Journal of Dermatology 40, 147–152 (1999). https://doi.org/10.1046/j.1440-0960.1999.00348.x

7. GLOSSARY OF TERMS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – from page 158

8. LIST OF ABBREVIATIONS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – from page 158